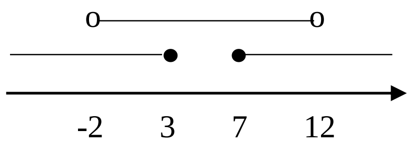
SM015 MATHEMATICS 1
SESSION 2025/2026
ANSWER SCHEME – PKA : PECUTAN PSPM

No	Section A
1 (a)	$\begin{aligned}\lim_{x\to -2}\frac{x^3+8}{x^2-4} &= \lim_{x\to -2}\frac{(x+2)(x^2-2x+4)}{(x+2)(x-2)}\\ &= \lim_{x\to -2}\frac{x^2-2x+4}{x-2}\\ &= \frac{(-2)^2-2(-2)+4}{-2-2}\\ &= -3\end{aligned}$
1 (b)	$\begin{aligned}\lim_{x\to 2}\frac{x-2}{3-\sqrt{x^2+5}} &= \lim_{x\to 2}\frac{x-2}{3-\sqrt{x^2+5}}\times\frac{3+\sqrt{x^2+5}}{3+\sqrt{x^2+5}}\\ &= \lim_{x\to 2}\frac{(x-2)\left(3+\sqrt{x^2+5}\right)}{9-(x^2+5)}\\ &= \lim_{x\to 2}\frac{(x-2)\left(3+\sqrt{x^2+5}\right)}{4-x^2}\\ &= \lim_{x\to 2}\frac{-(2-x)\left(3+\sqrt{x^2+5}\right)}{(2-x)(2+x)}\\ &= \lim_{x\to 2}\frac{-\left(3+\sqrt{x^2+5}\right)}{2+x}\\ &= \frac{-\left(3+\sqrt{2^2+5}\right)}{2+2}\\ &= -\frac{3}{2}\end{aligned}$
2	$y=\frac{x}{(x+2)^2}=\frac{A}{x+2}+\frac{B}{(x+2)^2}$ $\Rightarrow x=A(x+2)+B$ $x=-2:\ -2=B$, $B=-2$ Compare $x:\ 1=A$, $A=1$ $\begin{aligned}y=\frac{x}{(x+2)^2} &= \frac{1}{x+2}-\frac{2}{(x+2)^2}\\ &= (x+2)^{-1}-2(x+2)^{-2}\end{aligned}$

	$\frac{dy}{dx} = -(x+2)^{-2} + 4(x+2)^{-3}$ $= -\frac{1}{(x+2)^2} + \frac{4}{(x+2)^3}$ $\frac{d^2y}{dx^2} = 2(x+2)^{-3} - 12(x+2)^{-4}$ $= \frac{2}{(x+2)^3} - \frac{12}{(x+2)^4}$ When $x = 0$: $\frac{d^2y}{dx^2} = \frac{2}{2^3} - \frac{12}{2^4}$ $= -\frac{1}{2}$
3 (a)	$y = 3x^4 - 4x^3$ $\frac{dy}{dx} = 12x^3 - 12x^2$ $= 12x^2(x-1)$ $= 12x^2(x-1)$ $\frac{dy}{dx} = 0 \Rightarrow 12x^2(x-1) = 0$ $x = 0, 1$ Critical number $\frac{d^2y}{dx^2} = 36x^2 - 24x$ $= 12x(3x-2)$ When $x = 0$: $x < 0$, $\frac{d^2y}{dx^2} > 0$ $x > 0$, $\frac{d^2y}{dx^2} < 0$ When $x = 1$: $x < 1$, $\frac{d^2y}{dx^2} > 0$ $x > 1$, $\frac{d^2y}{dx^2} > 0$ $x = 0, y = 0 \Rightarrow (0,0)$ inflection point

No	Section B
1	$\frac{z}{z+2}=2-i$ $z=2z-zi+4-2i$ $z(1-i)=-4+2i$ $z=\frac{-4+2i}{1-i}$ $=\frac{-4+2i}{1-i}\times\frac{1+i}{1+i}$ $=\frac{-4-4i+2i+2i^2}{1-i^2}$ $=\frac{-6-2i}{2}$ $=-3-i$ $\bar{z}=-3+i$
2 (a)	$5\times5^{\log x}+5^{2-\log x}=30$ $5\left(5^{\log x}\right)+\frac{5^2}{5^{\log x}}=30$ Let $y=5^{\log x}$ $\Rightarrow 5y+\frac{25}{y}=30$ $5y^2+25=30y$ $5y^2-30y+25=0$ $y^2-6y+5=0$ $(y-1)(y-5)=0$ $y=1$, $y=5$ $5^{\log x}=5^0$, $5^{\log x}=5^1$ $\log x=0$, $\log x=1$ $x=1$ $\quad$ $x=10$
2 (b)	$2\le\|5-x\|<7$ $\|5-x\|\ge2$ and $\|5-x\|<7$ $5-x\le-2$ or $5-x\ge2$ $\quad$ $-7<5-x<7$ $x\ge7$ or $x\le3$ $\quad$ $-12<-x<2$ $-2<x<12$

	 Solution = $(-2,3]\cup[7,12)$
3 (a)	$f(x)=x^2+2kx+4$ $\quad = (x+k)^2-k^2+4$ $R_f=[4-k^2,\infty)$
3 (b)	$(f\circ g)(2)=4$ $f\left(g(2)\right)=4$ $f\left(3-k(2)\right)=4$ $[(3-2k)+k]^2-k^2+4=4$ $(3-k)^2-k^2$ $9-6k=0$ $k=\dfrac{3}{2}$
3 (c)	$(h\circ g)(x)=\dfrac{2}{8-3x}$ $h\left(g(x)\right)=\dfrac{2}{8-3x}$ $h\left(3-\dfrac{3}{2}x\right)=\dfrac{2}{8-3x}$ Let $y=3-\dfrac{3}{2}x$ $\dfrac{3}{2}x=3-y$ $x=\dfrac{2(3-y)}{3}$ $\Rightarrow h\left(y\right)=\dfrac{2}{8-3\left(\dfrac{2(3-y)}{3}\right)}=\dfrac{1}{1+y}$ $h(x)=\dfrac{1}{1+x}$

4 (a)	(ii) $(f \circ g)(x)$ $= f\left(g(x)\right)$ $= \frac{1}{2}\left[e^{4-[4-\ln(2x-3)]}+3\right]$ $= \frac{1}{2}\left[e^{\ln(2x-3)}+3\right]$ $= \frac{1}{2}\left(2x-3+3\right)$ $= \frac{2x}{2}$ $= x$ $(g \circ f)(x)$ $= g\left(f(x)\right)$ $= 4-\ln\left[2\left(\frac{1}{2}(e^{4-x}+3\right)-3\right]$ $= 4-\ln\left[e^{4-x}+3-3\right]$ $= 4-\ln e^{4-x}$ $= 4-(4-x)$ $= x$ $\therefore\ f^{-1}(x)=g(x)=4-\ln(2x-3)$ $D_{f^{-1}}=\left(\frac{3}{2},\infty\right)$, $R_{f^{-1}}=(-\infty,\infty)$
4 (b)	$R_f=[2,4]$ $2=m+\cos n(2\pi)$ …… (1) $4=m+\cos n(0)$ …… (2) From (2) : $4 = m + 1$ $m = 3$ From (1) : $2 = 3 + \cos n(2\pi)$ $\cos n(2\pi)=-1$ $n(2\pi)=\pi$ $n=\frac{1}{2}$
5 (a)	$\lim\limits_{x\to-\infty}\frac{\sqrt{x+4x^2}}{3x-1} = \lim\limits_{x\to-\infty}\frac{-\sqrt{\frac{x}{x^2}+\frac{4x^2}{x^2}}}{\frac{3x}{x}-\frac{1}{x}}$ $= \lim\limits_{x\to-\infty}\frac{-\sqrt{\frac{1}{x}+4}}{3-\frac{1}{x}}$ $= -\frac{2}{3}$

5 (b)	$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = f(2)$ $\lim_{x\to 2^-} \frac{\lvert x-2\rvert}{x-2} = \lim_{x\to 2^+}(ax^2 - bx + 3)$ $\lim_{x\to 2^-} \frac{-(x-2)}{x-2} = a(2)^2 - b(2) + 3$ $-1 = 4a - 2b + 3$ $4a - 2b = -4$ $2a - b = -2$ ..... (1) $\lim_{x\to 3^-} f(x) = \lim_{x\to 3^+} f(x) = f(3)$ $\lim_{x\to 3^-}(ax^2 - bx + 3) = \lim_{x\to 3^+}(2x - a + b)$ $a(3)^2 - b(3) + 3 = 2(3) - a + b$ $9a - 3b + 3 = 6 - a + b$ $10a - 4b = 3$ ..... (2) $(2) - 4(1): 2a = 11$ $a = \frac{11}{2}$ From (1) : $2\left(\frac{11}{2}\right) - b = -2$ $b = 13$
5c	$f(x) = \frac{x^2}{(x-1)(x-5)}$ $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \frac{x^2}{(x-1)(x-5)} = +\infty$ $\lim_{x\to 5^-} f(x) = \lim_{x\to 5^-} \frac{x^2}{(x-1)(x-5)} = -\infty$ $\therefore$ $x = 1$ or $x = 5$ are vertical asymptotes. $\lim_{x\to -\infty} \frac{x^2}{(x-1)(x-5)} = 1$ $\lim_{x\to +\infty} \frac{x^2}{(x-1)(x-5)} = 1$ $\therefore$ $y = 1$ is a horizontal asymptote.

6a	$y^4 = e^{x-y}$ (i) $4y^3 \dfrac{dy}{dx} = e^{x-y}\left(1-\dfrac{dy}{dx}\right)$ $= e^{x-y} - e^{x-y}\dfrac{dy}{dx}$ $= y^4 - y^4\dfrac{dy}{dx}$ $4y^3\dfrac{dy}{dx} + y^4\dfrac{dy}{dx} = y^4$ $y^3\dfrac{dy}{dx}(4+y) = y^4$ $\dfrac{dy}{dx} = \dfrac{y}{4+y}$ (ii) $\dfrac{d^2y}{dx^2} = \dfrac{(4+y)\dfrac{dy}{dx} - y\dfrac{dy}{dx}}{(4+y)^2}$ $(4+y)^2\dfrac{d^2y}{dx^2} = 4\dfrac{dy}{dx}$ $= 4\left(\dfrac{y}{4+y}\right)$ $(4+y)^3\dfrac{d^2y}{dx^2} = 4y$ $(4+y)^3\dfrac{d^2y}{dx^2} - 4y = 0$
6b	$x = \sqrt{\sin 2t} = (\sin 2t)^{\frac{1}{2}}$, $y = \sqrt{\cos 2t} = (\cos 2t)^{\frac{1}{2}}$ $\dfrac{dx}{dt} = \dfrac{1}{2}(\sin 2t)^{-\frac{1}{2}}.(\cos 2t.2)$ $\dfrac{dy}{dt} = \dfrac{1}{2}(\cos 2t)^{-\frac{1}{2}}.(-\sin 2t.2)$ $= \dfrac{\cos 2t}{\sqrt{\sin 2t}}$ $= -\dfrac{\sin 2t}{\sqrt{\cos 2t}}$ $\dfrac{dy}{dx} = \dfrac{dy}{dt}\times\dfrac{dt}{dx}$ $= -\dfrac{\sin 2t}{\sqrt{\cos 2t}}\times\left(\dfrac{\sqrt{\sin 2t}}{\cos 2t}\right)$ $= -\tan^{\frac{3}{2}} 2t$

	$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(-\tan 2t\right)^{\frac{3}{2}} \times \frac{dt}{dx}$ $= \left(-\frac{3}{2}(\tan 2t)^{\frac{1}{2}} \sec^2 2t.2\right) \times -\left(\tan 2t\right)^{\frac{1}{2}}.$ $= 3\tan 2t \sec^2 2t$
7a	$A = 400$ $xy = 400$ $y = \frac{400}{x}$ $C = x + y + y + \frac{1}{2}x$ $= \frac{3}{2}x + 2y$ $= \frac{3}{2}x + 2\left(\frac{400}{x}\right)$ $= \frac{3}{2}x + \frac{800}{x}$ $\frac{dC}{dx} = \frac{3}{2} - \frac{800}{x^2} = 0$ $\frac{800}{x^2} = \frac{3}{2}$ $x^2 = \frac{1600}{3}$ $x = \frac{40}{\sqrt{3}}$ $\frac{d^2C}{dx^2} = \frac{1600}{x^3}$ When $x = \frac{40}{\sqrt{3}}$, $\frac{d^2C}{dx^2} = \frac{1600}{\left(\frac{40}{\sqrt{3}}\right)^3} = 0.13 > 0$ (minimum) $y = \frac{400}{x} = \frac{400}{\left(\frac{40}{\sqrt{3}}\right)} = 10\sqrt{3}$ The cost is minimum when $x = \frac{40}{\sqrt{3}}$ m and $y = 10\sqrt{3}$ m.